

AULTBEA, United Kingdom

WMO: 030340

Lat: 57.8589N

Lon: 5.6328W

Elev: 33

StdP: 14.68

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	29.4	31.6	22.6	17.0	35.3	24.9	19.0	35.6	37.9	44.9	34.6	46.2	4.2	100	0.688

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	69.8	60.9	66.7	59.4	64.4	58.0	62.7	67.9	60.7	65.0	59.1	63.1	9.1	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.2	78.1	65.0	58.5	73.4	62.9	57.2	69.8	61.1	28.3	67.9	26.9	65.0	25.8	63.3	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.1	28.2	24.9	26.2	76.6	2.3	3.7	24.5	79.3	23.2	81.5	21.9	83.6	20.3	86.3		
(5)			25.1	66.1	2.3	3.6	23.5	68.7	22.2	70.8	20.9	72.8	19.2	75.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.8	41.8	41.6	43.1	46.3	50.0	54.2	57.4	57.3	54.9	50.4	45.7	42.7
	DBStd	7.24	4.66	4.53	4.39	4.62	4.63	3.78	3.69	3.38	3.54	4.15	4.73	5.27
	HDD50	1317	257	237	216	126	55	5	0	0	4	46	140	231
	HDD65	5909	721	655	679	560	464	325	237	240	303	454	579	692
	CDD50	890	1	1	2	17	57	130	231	228	151	57	11	4
	CDD65	6	0	0	0	0	0	0	3	3	0	0	0	0
	CDH74	21	0	0	0	0	2	1	9	9	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	11.1	13.3	13.3	12.4	10.5	9.6	9.4	8.7	9.2	10.1	11.6	12.1	12.8
(15) Precipitation	PrecAvg	61.7	7.1	6.3	5.0	3.5	3.6	3.1	3.4	4.3	5.4	6.5	7.0	6.6
	PrecMax	85.6	12.2	14.9	13.2	5.2	8.5	5.6	5.9	10.1	9.4	10.2	10.9	11.0
	PrecMin	43.1	1.2	2.2	1.0	1.6	0.7	1.1	1.0	1.8	1.8	2.2	2.6	1.2
	PrecStd	9.9	2.9	3.1	2.3	0.9	1.6	1.2	1.1	2.1	2.0	2.3	2.2	2.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.7	55.0	57.4	64.8	70.6	71.1	74.4	74.5	68.9	63.4	58.6	56.1
		MCWB	49.1	50.0	49.3	53.6	57.8	62.0	63.8	63.5	59.4	56.2	53.0	51.8
	2%	DB	51.5	51.8	52.7	58.9	64.5	66.3	69.5	68.5	65.0	60.1	55.5	53.6
		MCWB	48.8	48.9	48.0	51.2	54.4	58.9	62.0	61.1	58.0	54.9	51.7	50.7
	5%	DB	50.1	50.2	50.7	55.7	60.5	63.1	66.2	65.4	62.6	58.1	54.0	52.1
		MCWB	47.9	47.7	47.4	49.5	53.1	56.8	60.1	59.3	57.0	53.9	50.8	49.7
	10%	DB	48.5	48.5	49.3	53.4	57.5	60.3	63.6	63.0	60.5	56.4	52.5	50.7
		MCWB	46.1	46.1	46.5	48.4	51.5	55.1	58.4	57.9	56.0	52.7	49.7	48.2
	0.4%	WB	51.5	51.3	51.1	54.8	59.4	63.1	65.8	64.8	61.4	57.9	55.0	53.1
		MCDB	52.5	52.7	54.5	61.0	68.8	69.9	72.0	71.9	65.3	61.0	57.4	54.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	49.7	49.7	49.4	52.5	56.1	59.7	63.1	62.1	59.4	56.2	52.9	51.4
		MCDB	50.8	51.3	51.3	57.2	62.1	64.7	68.2	67.0	63.2	59.0	54.6	53.0
	5%	WB	48.1	48.2	48.2	50.7	54.0	57.7	60.9	60.3	57.9	54.8	51.3	50.2
		MCDB	49.8	49.9	50.1	54.3	59.0	61.8	64.8	64.4	61.3	57.4	53.3	52.0
	10%	WB	46.6	46.4	47.0	49.2	52.2	55.9	59.2	58.5	56.7	53.3	49.9	48.3
		MCDB	48.2	48.3	49.1	52.5	56.6	59.3	62.9	62.0	59.7	55.9	52.3	50.5
	5% DB	MDBR	7.4	8.0	8.8	10.4	11.3	10.0	9.5	9.3	9.3	8.5	7.8	7.9
		MCDBR	7.5	8.7	10.4	14.4	17.2	15.2	14.6	14.0	12.8	10.6	8.5	8.8
		MCWBR	7.4	7.3	7.5	8.9	10.1	9.3	8.5	8.2	8.1	7.6	7.5	8.6
		MCWBR	7.4	7.4	8.8	12.0	15.1	13.5	13.4	12.9	10.9	9.0	8.2	8.5
(35) Mean Daily Temperature Range	5% WB	MCWBR	7.7	7.0	7.2	8.2	9.6	9.2	8.4	8.3	7.8	7.7	8.0	8.7
(40) Clear-Sky Solar Irradiance	taub		0.341	0.352	0.369	0.384	0.385	0.376	0.380	0.369	0.363	0.347	0.328	0.323
	taud		2.136	2.241	2.280	2.297	2.349	2.413	2.418	2.455	2.449	2.371	2.218	2.103
	Ebn at Noon		148	211	244	261	270	273	270	265	247	213	155	116
	Edh at Noon		18	24	30	35	36	34	33	30	26	21	16	13
(44) All-Sky Solar Radiation	RadAvg		112	301	632	1085	1459	1435	1341	1075	733	380	153	77
	RadStd		16	37	51	77	109	138	117	67	51	39	15	9

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page